

Mostafa Hassan Al-Sayed Muhammad Shaheen

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PROFESSIONAL SUMMARY

Artificial Intelligence Engineer with practical experience in machine learning and computer vision. Develops full-stack applications and data solutions using TensorFlow and PyTorch. Integrates AI models into web platforms and cross-platform mobile apps (Flutter). Uses AI-assisted tools (Vibe Coding) and builds Autonomous AI Agents to speed up development workflows. Focuses on turning complex AI concepts into functional, user-centric applications.

CORE COMPETENCIES & SKILLS

- **AI & Machine Learning:** Autonomous AI Agents, Feature Engineering, Data Engineering, Computer Vision, Deep Learning, TensorFlow, PyTorch, NLP, Predictive Modeling.
- **Modern Development:** AI-Assisted Coding (Vibe Coding), Generative AI Tooling, LLM Integration, Rapid Prototyping.
- **Software & Web Development:** Full-Stack Web Development, Python, C++, HTML/CSS, JavaScript, RESTful APIs, SQL.
- **Mobile App Development:** Flutter, Dart, Cross-platform Architecture, State Management, Mobile UI/UX.
- **Digital Marketing & SEO:** Search Engine Optimization (SEO), Technical SEO, Web Analytics.
- **Tools & Methodologies:** Git, Agile Development, GUI Development (PyQt, Kivy), Secure Software Practices.

PROFESSIONAL EXPERIENCE

Technical Trainer & Mentor ("She Leads" National Initiative)

Feb 2026 - Jul 2026

Youth Leaders Foundation (YLF) | Supported by Telecom Egypt & Banque Misr

- Served as the primary Technical Trainer for 2 engineering teams, guiding them from ideation to prototype deployment under the patronage of the Ministry of Education.
- Directed the "Eye Guide" team in building functional hardware-software prototypes for the visually impaired, linked to the "Baseet" companion mobile app.
- Led the "Eye Guide" team to win 1st Place Nationwide in the Manufacturing category, securing a financial grant for the product.
- Supervised a second team in designing and developing an AI-integrated artificial bionic arm.

Artificial Intelligence & Machine Learning Trainee

Dec 2024 - Jan 2025

Omal Misr Complex - Cairo, Egypt

- Completed an intensive 2-month training program focused on applying Machine Learning algorithms and Computer Vision applications.
- Engineered predictive neural networks using TensorFlow and PyTorch to process datasets of over 10,000 records, achieving 90% accuracy.
- Deployed AI models that optimized data processing pipelines, reducing manual analysis time by an estimated 30%.

Software Developer & Technical SEO (Freelance)

Jan 2023 - Present

Independent Projects & Digital Publications - Cairo, Egypt

- Developed and deployed responsive web platforms and high-performance Flutter mobile apps for 3+ independent clients.
- Managed SEO strategies and content optimization for a digital news publication to improve site indexing and online visibility.
- Enforced technical best practices to optimize page load speeds and proper organic search presence.

EDUCATION

Bachelor of Science in Artificial Intelligence, Capital International Technology University

Expected Graduation: 2028

Technology Faculty - Cairo, Egypt (Formerly Helwan International Technology University)

- **Current Grade:** Excellent (Cumulative)
- **Relevant Coursework:** Machine Learning, Data Structures, Algorithms, Computer Vision, Software Engineering, Database Systems.
- **Leadership:** Spearheaded academic initiatives as Cohort Leader for 100+ students, acting as the primary liaison with faculty.

ACADEMIC & KEY PROJECTS

"Memo" - Autonomous Local Voice Assistant (Edge AI)

2026

Hybrid Minds Team | Capital International Technology University

- Engineered a fully offline, Arabic-speaking AI voice assistant running locally on a Raspberry Pi 5 using local LLMs (Ollama, Qwen) and custom ONNX wake-word models.
- Architected a distributed system over LAN with a lightweight Flask agent and MQTT fallback, executing 20+ custom callable tools for PC automation.
- Developed a custom text-stream parser and implemented sentence-level parallelism for streaming Edge-TTS, achieving near-zero latency between LLM reasoning and speech.

Smart Contactless Elevator (SCEM) - IoT & Predictive Maintenance

2025

Hardware Simulation & Mobile App Dashboard

- Designed an IoT contactless elevator panel to minimize disease transmission, featuring automated audio feedback systems for visually impaired accessibility.
- Developed a companion mobile application providing real-time data monitoring, remote contactless control, and predictive maintenance analytics.